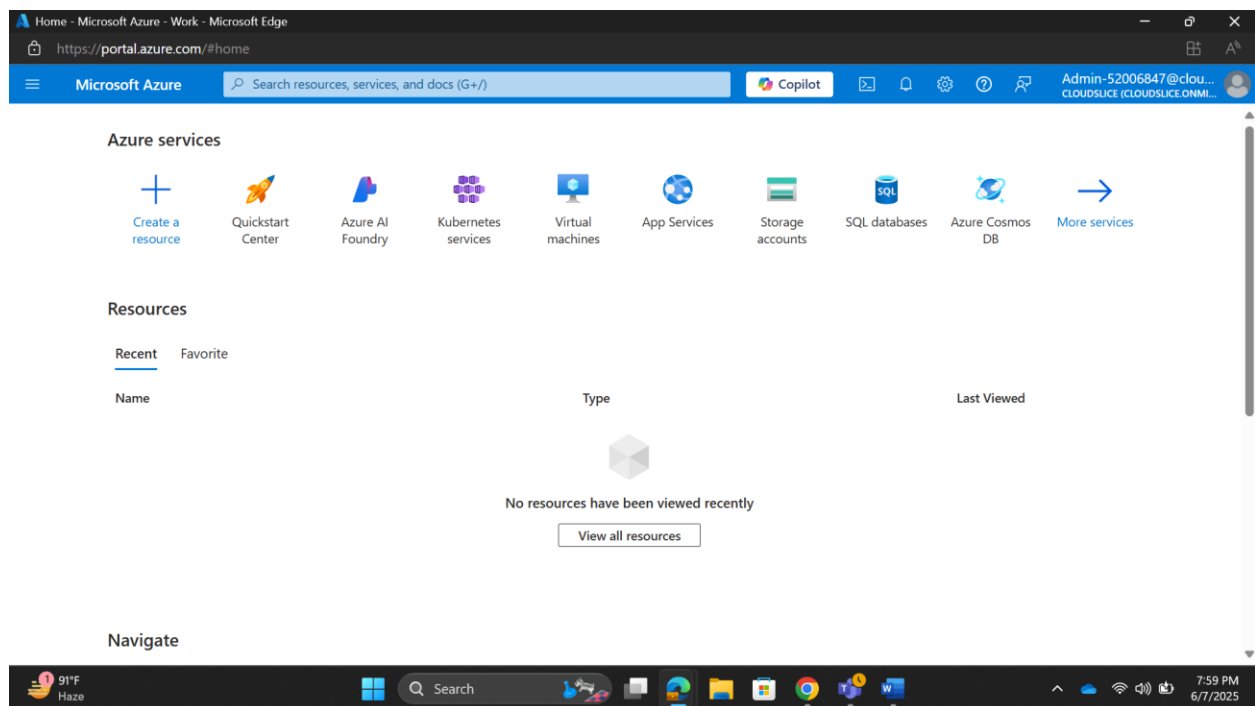


Lab 3 Configure Route Tables in a Virtual Network

In this challenge, you will ensure that all traffic between the front-end and back-end subnets of the virtual network are routed through a firewall virtual appliance. First, you will create the route tables and routes, and then you will then associate them to the subnets. Next, you will enable IP forwarding on the virtual appliance. Finally, you will verify that traffic is being routed through the virtual appliance. Note: Once you begin the challenge, you will not be able to pause, save, or exit and then return to your challenge. Please ensure that you have set aside enough time to complete the challenge before you start.

Task 1: Record the IP addresses of existing virtual machines

Login to Microsoft Azure with given credentials



Compute infrastructure - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#view/Microsoft_Azure_ComputeHub/ComputeHubMenuBlade/~/_virtualMachinesBrowse

Microsoft Azure Search resources, services, and docs (G+/J) Copilot Admin-52006847@clou... CLOUDSLICE (CLOUDSLICE.ONMI...

Home > Compute infrastructure

Compute infrastructure | Virtual machines

Virtual machines Get started

Overview All resources Infrastructure Virtual machines Virtual Machine Scale Set (VMSS) Compute Fleet Disks + images Capacity + placement Related services Help

+ Create Switch to classic Reservations Manage view Refresh Export to CSV Group by none

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

Filter for any field... Subscription equals all Type equals all Resource Group equals all Location equals all Add filter

	Name ↑	Subscription	Resource Group	Location	Status	Operating syst...	Size
☐	VM1	Challenge Labs ...	RG1lod52006847	East US	Running	Linux	Standard_DS1_v2
☒	VM2	Challenge Labs ...	RG1lod52006847	East US	Running	Linux	Standard_DS1_v2
☐	VM3	Challenge Labs ...	RG1lod52006847	East US	Running	Linux	Standard_DS1_v2

Showing 1 - 3 of 3. Display count: 10 Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

91°F Haze Search VM2 - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a59a8733-537b-465e-a883-4b94bfbf1069/resourceGroups/RG1lod52006847/providers/Microsoft.Com...

Microsoft Azure Search resources, services, and docs (G+/J) Copilot Admin-52006847@clou... CLOUDSLICE (CLOUDSLICE.ONMI...

Home > Compute infrastructure | Virtual machines

VM2

Virtual machine

Search Help me copy this VM in any region

Connect Start Restart Stop Hibernate Capture Delete Refresh Open in mobile Feedback

Essentials

Resource group (move) [RG1lod52006847](#)

Status Running

Location East US

Subscription (move) [Challenge Labs 09](#)

Subscription ID a59a8733-537b-465e-a883-4b94bfbf1069

Tags (edit) [Add tags](#)

Operating system Linux (ubuntu 18.04)

Size Standard DS1 v2 (1 vcpu, 3.5 GiB memory)

Public IP address [4.236.129.149](#)

Virtual network/subnet [app-vnet/backend](#)

DNS name [Not configured](#)

Health state -

Time created 6/7/2025, 2:25 PM UTC

Add or remove favorites by pressing Ctrl+Shift+F

91°F Haze Search

Copy public IP address

Do the same for VM3

Task 2: Configure a front-end route table

The image shows two screenshots of the Microsoft Azure portal. The top screenshot displays the 'Route tables' page, which indicates that no route tables are currently displayed. It includes a 'Create' button and a 'Learn more' link. The bottom screenshot shows the 'Create Route table' wizard, specifically the 'Project details' section. In this section, the 'Subscription' is set to 'Challenge Labs 09', the 'Resource group' is 'RG1lod52006847', the 'Region' is 'East US', and the 'Name' is 'app-frontend-rt'. The 'Propagate gateway routes' option is set to 'No'. The 'Review + create' button is highlighted at the bottom of the wizard.

Route tables - Microsoft Azure - Work - Microsoft Edge
https://portal.azure.com/#browse/Microsoft.Network%2FrouteTables

Home > Route tables

Route tables

cloudslice (cloudslice.onmicrosoft.com)

+ Create Manage view Refresh Export to CSV Open query Assign tags Group by none

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

Filter for any field... Subscription equals all Resource Group equals all Location equals all Add filter

No route tables to display

Create a route table when you need to override Azure's default routing. For example, you can route traffic to a network virtual appliance or to your on-premises network for inspection. A route table contains routes and is associated to one or more subjects.

+ Create

Learn more

Showing 1 - 0 of 0. Display count: 10

Give feedback

Create Route table - Microsoft Azure - Work - Microsoft Edge
https://portal.azure.com/#create/Microsoft.RouteTable

Home > Route tables >

Create Route table

Basics Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Challenge Labs 09

Resource group * RG1lod52006847
Create new

Instance details

Region * East US

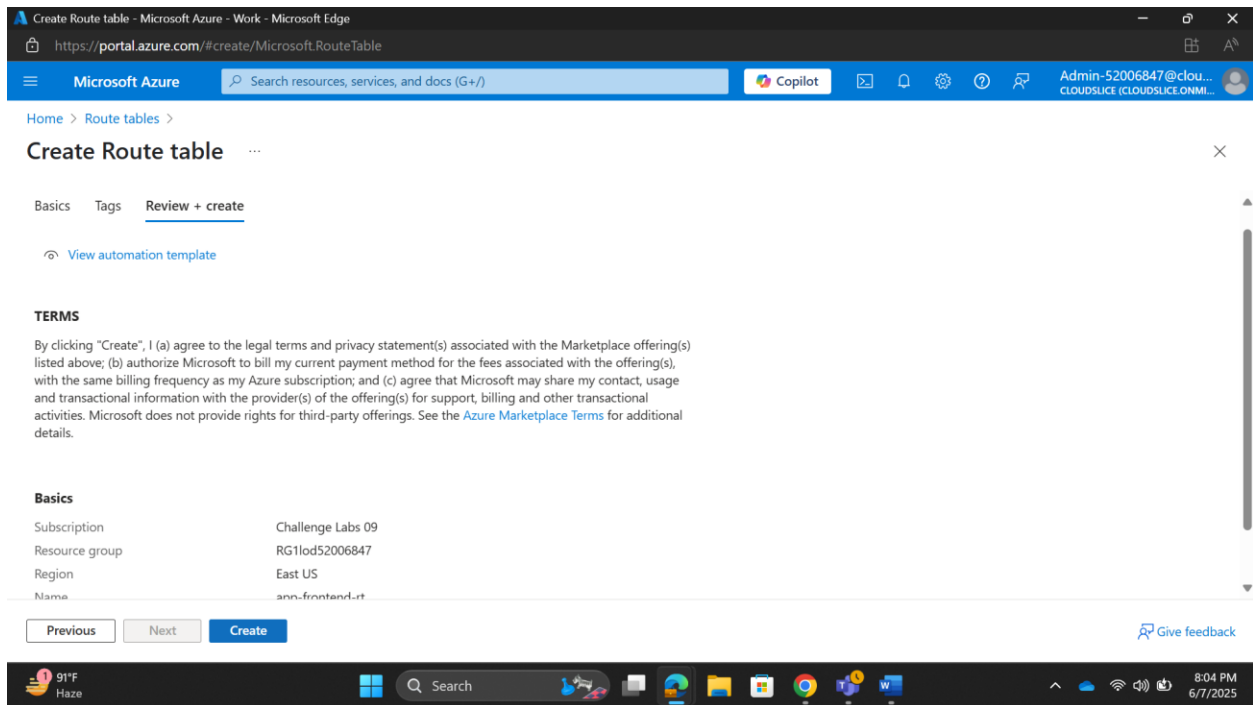
Name * app-frontend-rt

Propagate gateway routes * ☐ Yes ☒ No

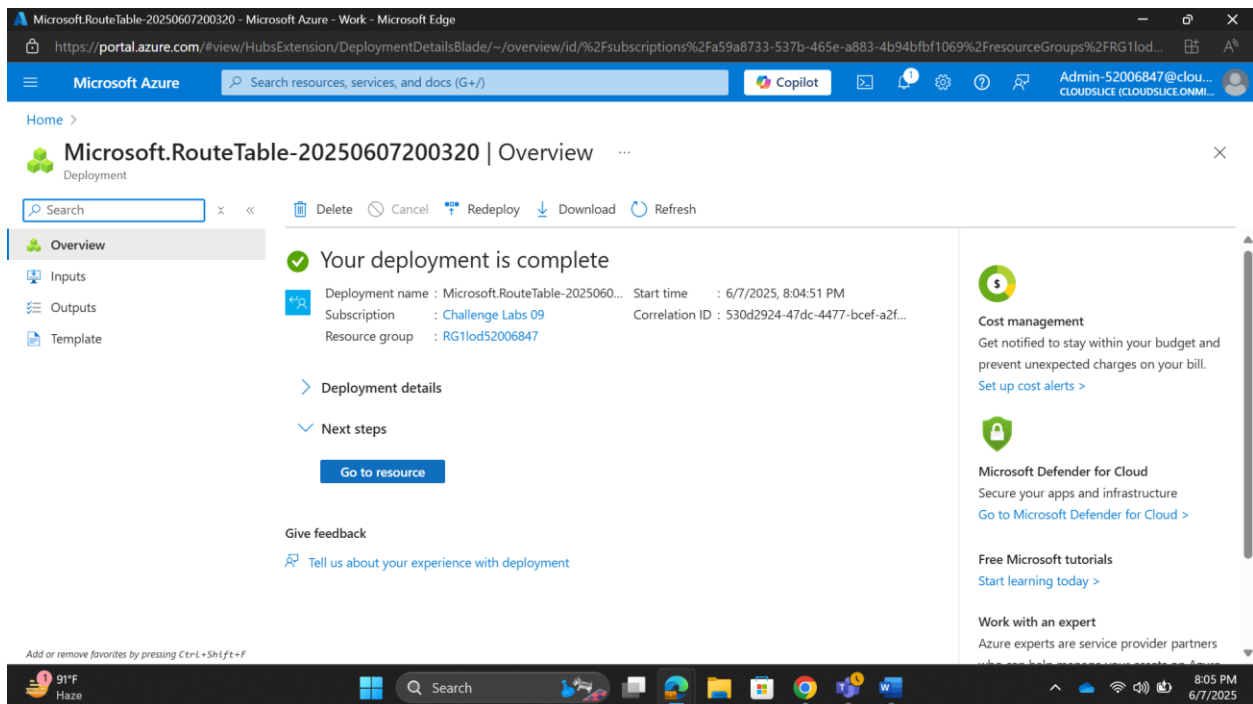
Previous Next Review + create

Give feedback

Click review + create



click create



click go to resources

app-frontend-rt - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a59a8733-537b-465e-a883-4b94bfbf1069/resourceGroups/RG1lod52006847/providers/Microsoft.Net...

Microsoft Azure Search resources, services, and docs (G+)

Admin-52006847@clou... CLOUDSLICE (CLOUDSLICE.ONMI...

Home > Microsoft.RouteTable-20250607200320 | Overview > RG1lod52006847 > app-frontend-rt

app-frontend-rt | Routes

Route table

Search

+ Add Refresh Give feedback

Search routes

Name ↑↓	Address prefix ↑↓	Next hop type ↑↓	Next hop IP address ↑↓
No results.			

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Configuration
Routes
Subnets
Properties
Locks
Monitoring

Add or remove favorites by pressing Ctrl+Shift+F

91°F
Haze

Search

8:06 PM
6/7/2025

Add route - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a59a8733-537b-465e-a883-4b94bfbf1069/resourceGroups/RG1lod52006847/providers/Microsoft.Net...

Microsoft Azure Search resources, services, and docs (G+)

Admin-52006847@clou... CLOUDSLICE (CLOUDSLICE.ONMI...

Home > Microsoft.RouteTable-20250607200320 | Overview > RG1lod52006847 > app-frontend-rt

app-frontend-rt | Routes

Route table

Search

+ Add Refresh Give feedback

Search routes

Name ↑↓	Address prefix ↑↓
No results.	

Add route

app-frontend-rt

A user defined route (UDR) is a static route that overrides Azure's default system routes, or adds a route to a subnet's route table. [Learn more](#)

Route name * to-backend ✓

Destination type * IP Addresses ✓

Destination IP addresses/CIDR ranges * 10.1.1.0/24 ✓

Next hop type * Virtual appliance ✓

Next hop address * 10.1.255.4 ✓

Ensure you have IP forwarding enabled on your virtual appliance. You can enable this by navigating to the respective network interface's IP address settings.

Add Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

91°F
Haze

Search

8:08 PM
6/7/2025

click add

app-frontend-rt - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a59a8733-537b-465e-a883-4b94bfb1069/resourceGroups/RG1lod52006847/providers/Microsoft.Net...

Microsoft Azure Search resources, services, and docs (G+/)

Copilot Admin-52006847@clou... CLOUDSLICE (CLOUDSLICE.ONMI...

Home > Microsoft.RouteTable-20250607200320 | Overview > RG1lod52006847 > app-frontend-rt

app-frontend-rt | Subnets

Route table

Search

+ Associate

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Configuration

Routes

Subnets

Properties

Locks

Monitoring

Search subnets

Name ↑↓	Address range ↑↓	Virtual network ↑↓	Security group ↑↓
No results.			

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

91°F Haze 8:08 PM 6/7/2025

Associate subnet - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a59a8733-537b-465e-a883-4b94bfb1069/resourceGroups/RG1lod52006847/providers/Microsoft.Net...

Microsoft Azure Search resources, services, and docs (G+/)

Copilot Admin-52006847@clou... CLOUDSLICE (CLOUDSLICE.ONMI...

Home > Microsoft.RouteTable-20250607200320 | Overview > RG1lod52006847 > app-frontend-rt

app-frontend-rt | Subnets

Route table

Search

+ Associate

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Configuration

Routes

Subnets

Properties

Locks

Monitoring

Search subnets

Name ↑↓	Address range ↑↓
No results.	

Associate subnet

app-frontend-rt

Virtual network ⓘ

app-vnet (RG1lod52006847)

Subnet * ⓘ

frontend

OK

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

91°F Haze 8:09 PM 6/7/2025

Click ok

[illegible]

Create Route table - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#create/Microsoft.RouteTable

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Admin-52006847@clou...
CLOUDSLICE (CLOUDSLIKE ONMI...

Home > Route tables >

Create Route table

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Challenge Labs 09

Resource group * ⓘ

RG1lod52006847

Create new

Instance details

Region * ⓘ

East US

Name * ⓘ

app-backend-rt

Propagate gateway routes * ⓘ

Yes

No

Previous

Next

Review + create

Give feedback

Create Route table - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#create/Microsoft.RouteTable

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Admin-52006847@clou...
CLOUDSLICE (CLOUDSLICE.ONMI...

Home > Route tables >

Create Route table

[view automation template](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Basics

Subscription	Challenge Labs 09
Resource group	RG1lod52006847
Region	East US
Name	app-backend-rt
Propagate gateway routes	No

Previous

Next

Create

[Give feedback](#)

91°F
Haze

Search

8:12 PM
6/7/2025

app-backend-rt - Microsoft Azure - Work - Microsoft Edge

https://portal.azure.com/#@cloudslice.onmicrosoft.com/resource/subscriptions/a59a8733-537b-465e-a883-4b94bfbf1069/resourceGroups/RG1lod52006847/providers/Microsoft.Net...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Admin-52006847@clou...
CLOUDSLICE (CLOUDSLICE.ONMI...

Home > Microsoft.RouteTable-20250607201108 | Overview > RG1lod52006847 > app-backend-rt

app-backend-rt | Routes

Route table

Search

+ Add Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Configuration

Routes

Subnets

Properties

Locks

Monitoring

Search routes

Name ↑↓	Address prefix ↑↓	Next hop type ↑↓	Next hop IP address ↑↓
No results.			

Add or remove favorites by pressing Ctrl+Shift+F

91°F
Haze

Search

8:12 PM
6/7/2025

Microsoft Azure portal interface showing the "Add route" configuration for the "app-backend-rt" route table. The left sidebar displays navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Configuration, Routes (selected), Subnets, Properties, Locks, and Monitoring. The main content area shows the "Add route" form with the following fields:

- Route name: to-frontend
- Destination type: IP Addresses
- Destination IP addresses/CIDR ranges: 10.1.0.0/24
- Next hop type: Virtual appliance
- Next hop address: 10.1.255.4

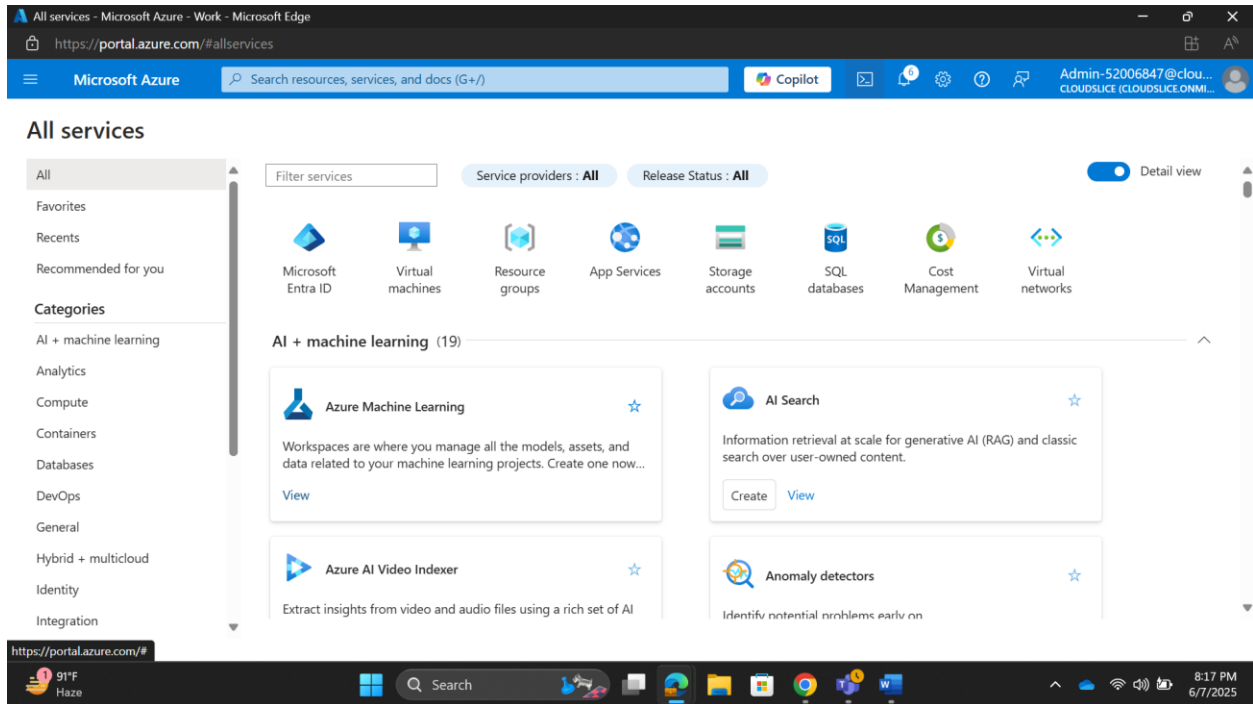
A note indicates: "Ensure you have IP forwarding enabled on your virtual appliance. You can enable this by navigating to the respective network interface's IP address settings." The "Add" button is visible at the bottom of the form.

Microsoft Azure portal interface showing the "Associate subnet" configuration for the "app-backend-rt" route table. The left sidebar displays navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Configuration, Routes, Subnets (selected), Properties, Locks, and Monitoring. The main content area shows the "Associate subnet" form with the following fields:

- Virtual network: app-vnet (RG1lod52006847)
- Subnet: backend

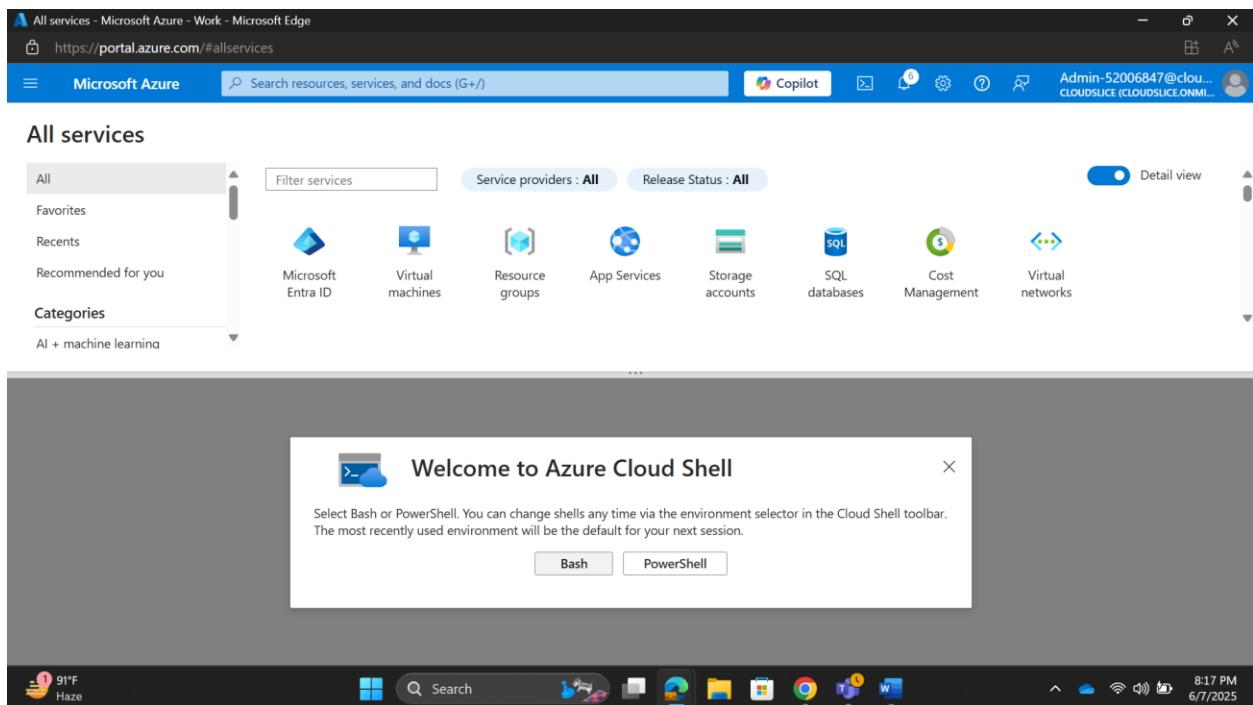
The "OK" button is visible at the bottom of the form.

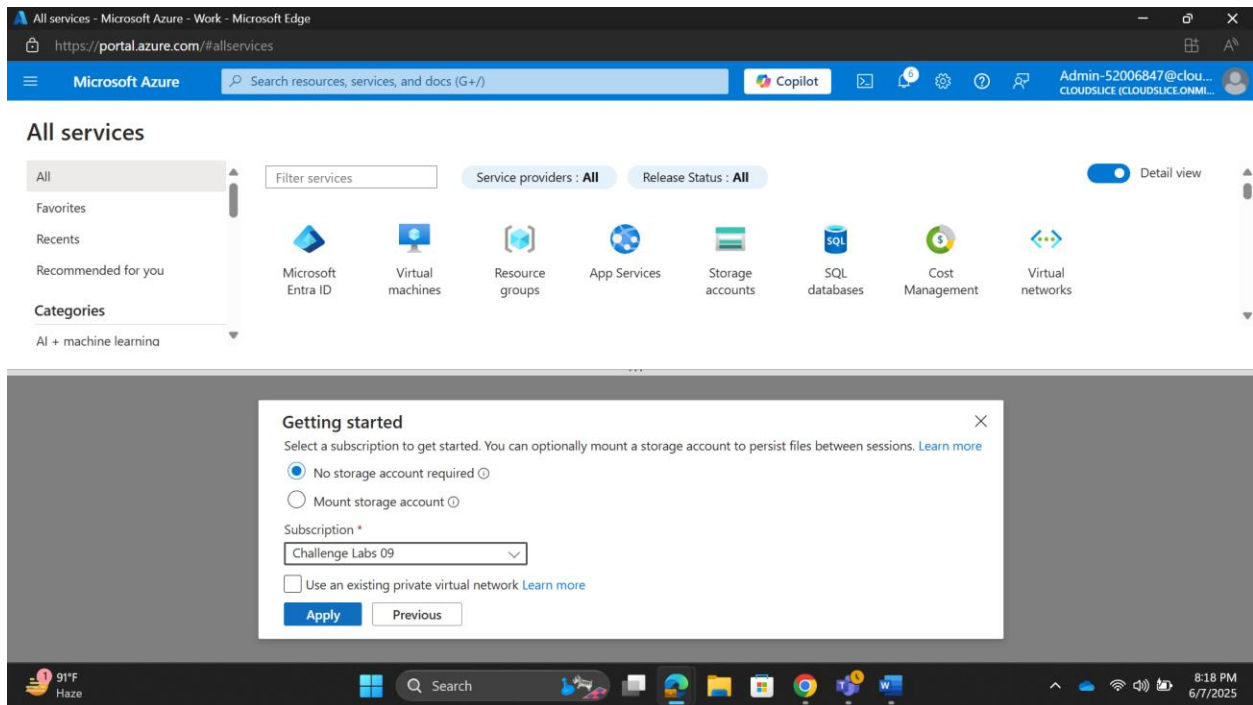
Task 4: Enable IP forwarding on the virtual appliance



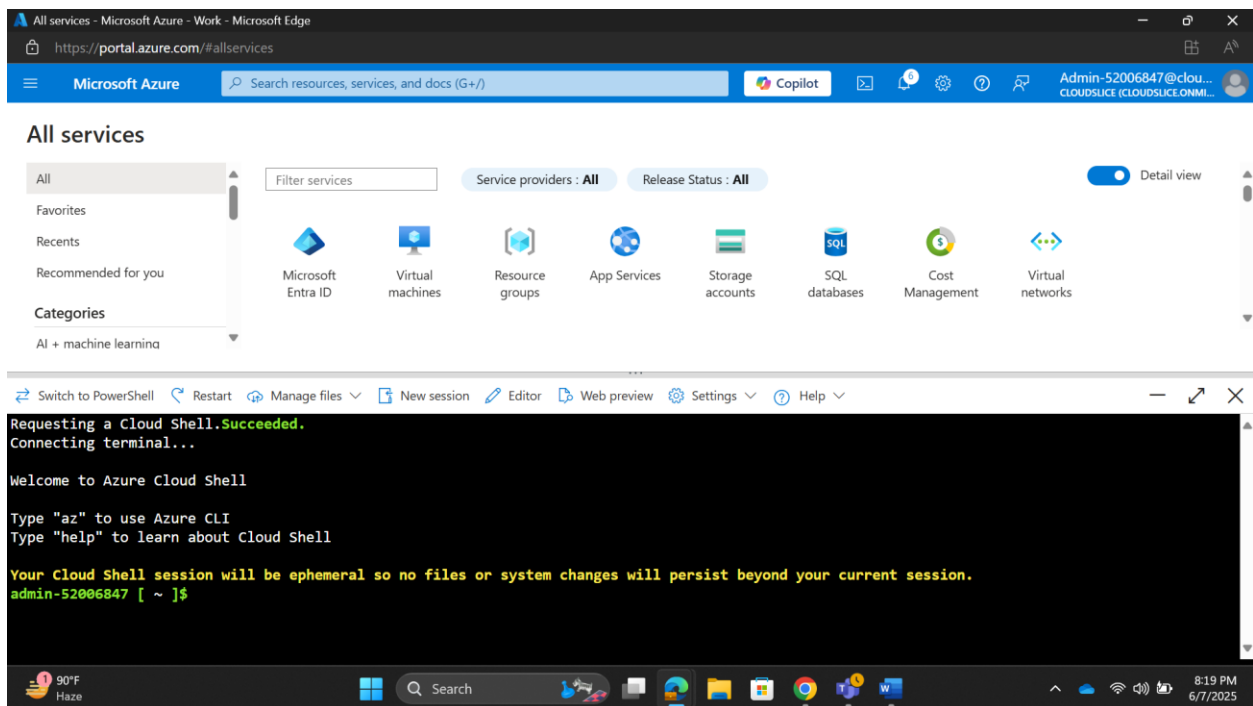
Click Cloud shell icon

Select bash

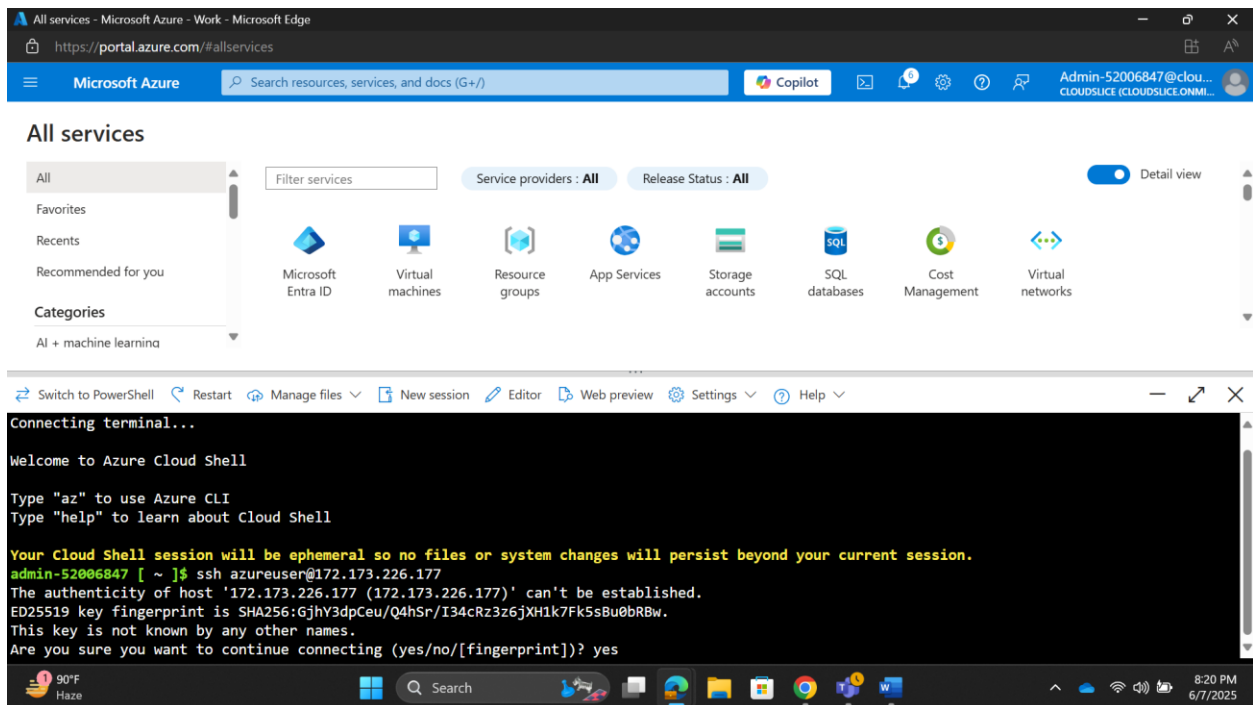




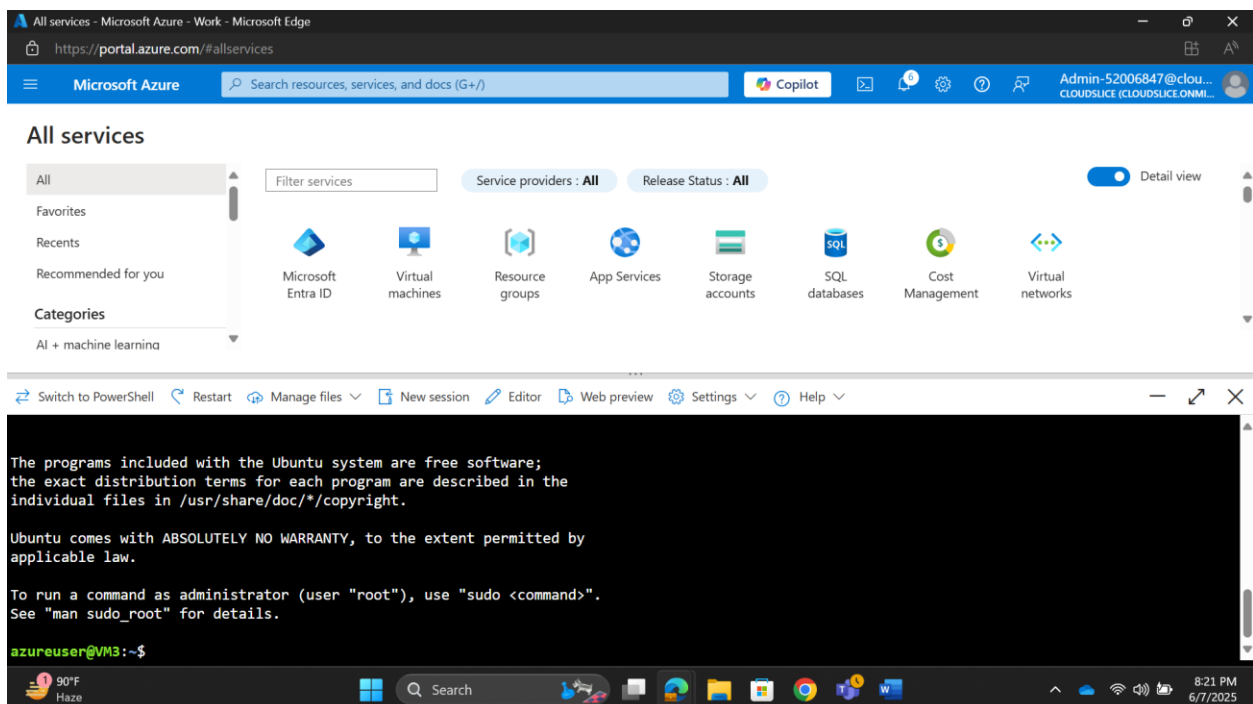
Click apply



Run command/ username: ssh azureuser@172.173.226.177

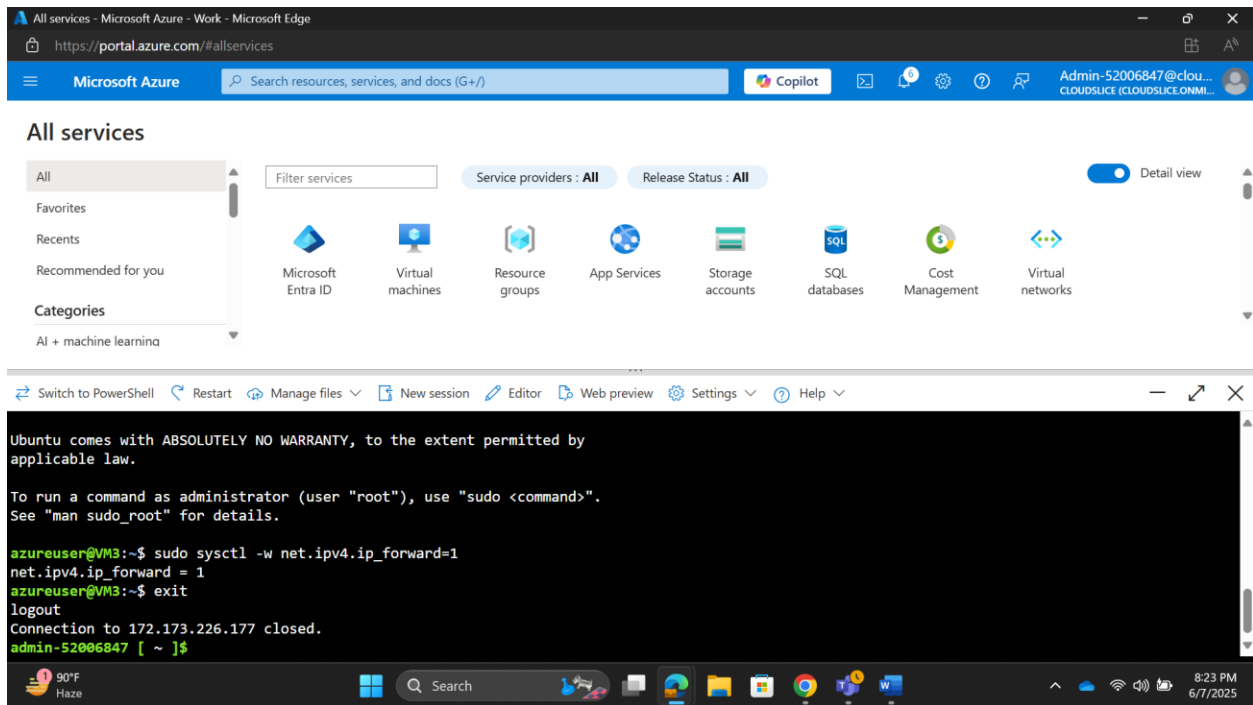


Enter password, click enter

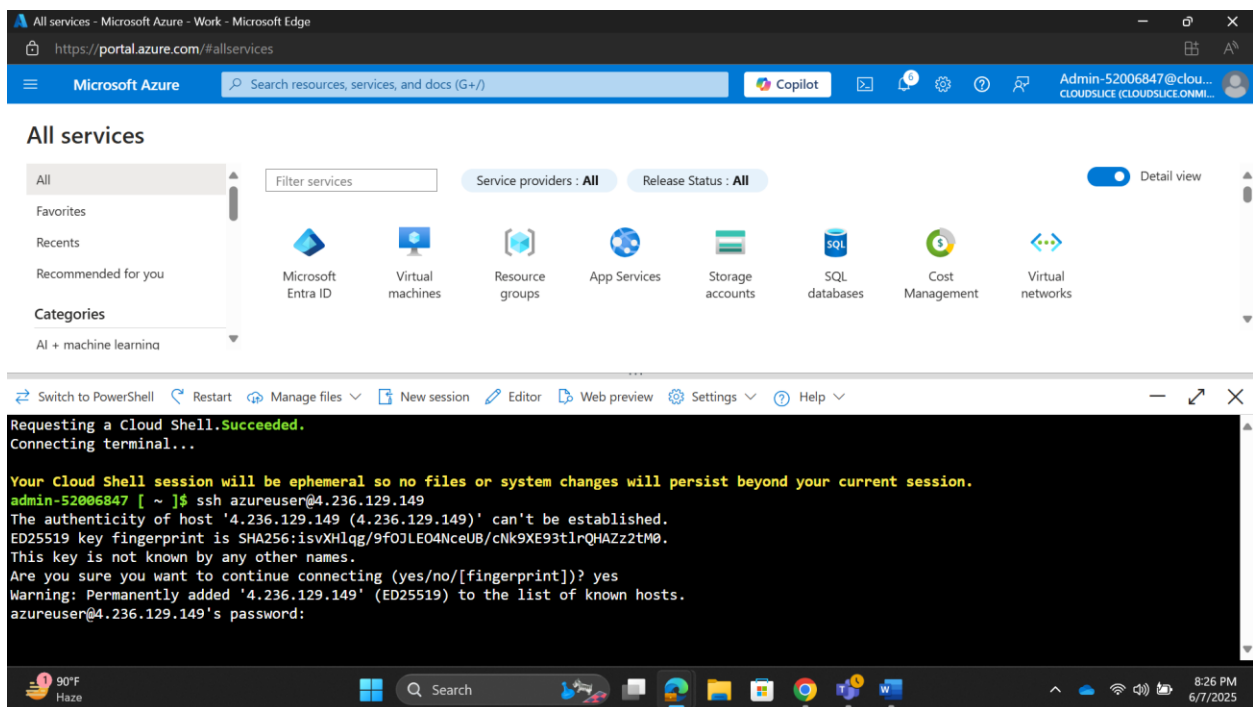


Enter command : sudo sysctl -w net.ipv4.ip_forward=1

Enter exit, click enter



Task 5: Test routes by using the traceroute tool



sudo apt-get update

```
All services - Microsoft Azure - Work - Microsoft Edge
https://portal.azure.com/#allservices

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Admin-52006847@clou...
CLOUDSLICE (CLOUDSLICE.ONMIL...)

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

azureuser@VM2:~$ sudo apt-get update
Hit:1 http://azure.archive.ubuntu.com/ubuntu bionic InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu bionic-updates InRelease [102 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu bionic-backports InRelease [102 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu bionic-security InRelease [102 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu bionic/universe amd64 Packages [8570 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu bionic/universe Translation-en [4941 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu bionic/multiverse amd64 Packages [151 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu bionic/multiverse Translation-en [108 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu bionic-updates/main amd64 Packages [3044 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu bionic-updates/main Translation-en [554 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu bionic-updates/universe amd64 Packages [1914 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu bionic-updates/universe Translation-en [421 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu bionic-updates/multiverse amd64 Packages [25.6 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu bionic-updates/multiverse Translation-en [6088 B]
Get:15 http://azure.archive.ubuntu.com/ubuntu bionic-backports/main amd64 Packages [53.3 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu bionic-backports/main Translation-en [14.6 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu bionic-backports/universe amd64 Packages [18.2 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu bionic-backports/universe Translation-en [8668 B]
Get:19 http://azure.archive.ubuntu.com/ubuntu bionic-security/main amd64 Packages [2717 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu bionic-security/main Translation-en [468 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu bionic-security/universe amd64 Packages [1303 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu bionic-security/universe Translation-en [308 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu bionic-security/multiverse amd64 Packages [19.8 kB]
Get:24 http://azure.archive.ubuntu.com/ubuntu bionic-security/multiverse Translation-en [3928 B]
Fetched 25.0 MB in 6s (4311 kB/s)
Reading package lists... Done
```

sudo apt install traceroute

```
All services - Microsoft Azure - Work - Microsoft Edge
https://portal.azure.com/#allservices

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Admin-52006847@clou...
CLOUDSLICE (CLOUDSLICE.ONMIL...)

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Get:24 http://azure.archive.ubuntu.com/ubuntu bionic-security/multiverse Translation-en [3928 B]
Fetched 25.0 MB in 6s (4311 kB/s)
Reading package lists... Done
azureuser@VM2:~$ sudo apt install traceroute
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 10 not upgraded.
Need to get 45.4 kB of archives.
After this operation, 152 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu bionic/universe amd64 traceroute amd64 1:2.1.0-2 [45.4 kB]
Fetched 45.4 kB in 0s (3865 kB/s)
Selecting previously unselected package traceroute.
(Reading database ... 59190 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.0-2_amd64.deb ...
Unpacking traceroute (1:2.1.0-2) ...
Setting up traceroute (1:2.1.0-2) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: using /usr/sbin/tcpttraceroute.db to provide /usr/sbin/tcpttraceroute (tcpttraceroute) in auto mode
Processing triggers for man-db (2.8.3-2ubuntu.0.1) ...
azureuser@VM2:~$
```

traceroute VM1

```
Microsoft Azure portal showing terminal output for installing traceroute on a VM. The output shows the successful installation of traceroute and the results of a traceroute command to VM1 (10.1.0.4).
```

Reading package lists... Done
azureuser@VM2:~\$ sudo apt install traceroute
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
traceroute
0 upgraded, 1 newly installed, 0 to remove and 10 not upgraded.
Need to get 45.4 kB of archives.
After this operation, 152 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu bionic/universe amd64 traceroute amd64 1:2.1.0-2 [45.4 kB]
Fetched 45.4 kB in 0s (3865 kB/s)
Selecting previously unselected package traceroute.
(Reading database ... 59190 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.0-2_amd64.deb ...
Unpacking traceroute (1:2.1.0-2) ...
Setting up traceroute (1:2.1.0-2) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
azureuser@VM2:~\$ traceroute VM1
traceroute to VM1 (10.1.0.4), 30 hops max, 60 byte packets
1 vm3.internal.cloudapp.net (10.1.255.4) 1.615 ms 1.597 ms 1.583 ms
2 * * vm1.internal.cloudapp.net (10.1.0.4) 2.988 ms
azureuser@VM2:~\$

Great job! [Print](#)

Your dedication and hard work really paid off!



Areas where you excelled

- **Recording the IP addresses of the existing virtual machines**
You showed your skills in this area and finished with success.
- **Configure a front-end route table**
You showed your skills in this area and finished with success.
- **Configure a back-end route table**
You showed your skills in this area and finished with success.
- **Enable IP forwarding on the virtual appliance**

You are done with the lab!!!